Feather: 3D sketchbook light as a feather

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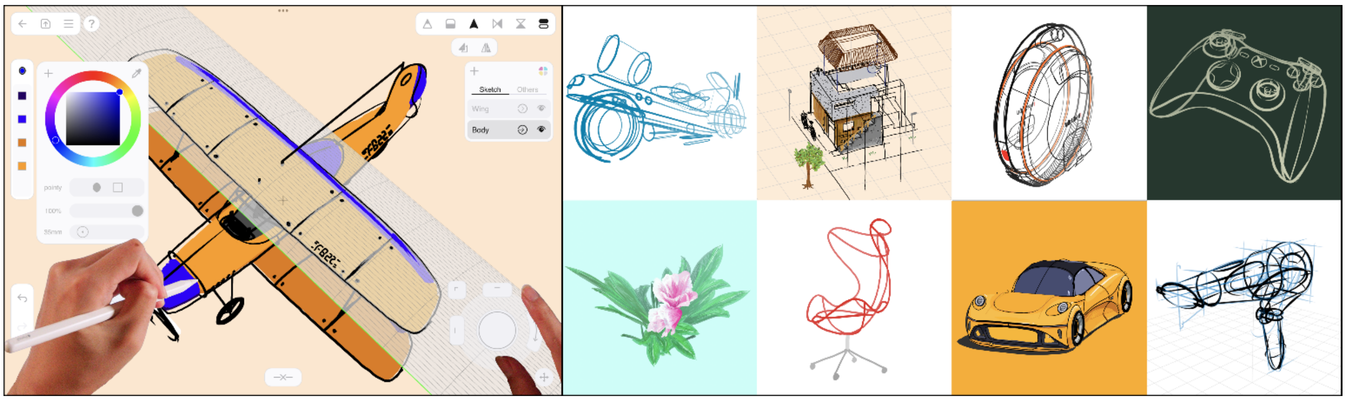

Figure 1: Feather is a pen & touch 3D curve drawing system to create various 3D sketches and artworks (www.feather.art).

ABSTRACT

We present Feather, a 3D drawing system that allows users to create 3D curve artworks with a pen & touch interface on a tablet. We implemented multi-view 3D sketching by applying a concept of drawing & bending a "3D guide" and a view-oriented joystick widget that streamlines tedious 3D transformations—all of which are integrated into a single web application while maintaining the usability and look of conventional 2D drawing software.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interactive systems and tools.

KEYWORDS

3D sketching, 3D drawing, 3D transformation, tablet, pen, touch

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1 INTRODUCTION

Advances in real-time graphics technology drive faster iteration cycles in the creative process. Increasing outlets for 3D content, such as virtual and mixed reality, expand the demand for 3D creation beyond the design and entertainment industries.

However, in the early stages of creation, traditional pen-and-paper 2D sketching remains irreplaceable for capturing ideas on a "blank slate." Since 2D sketches represent shapes from specific viewpoints, transitioning to 3D geometry is fraught with information gaps and inconsistencies. The limited ability to quickly express 3D shape ideas becomes a "bottleneck" for later iterations with advanced 3D graphics technologies.

We developed Feather, a web-based 3D sketching application for early 3D creation. Users can generate 3D curves from multiple viewpoints using a 2D pen & touch input (Figure 1). Feather offers a cross-device environment, allowing users to draw anywhere and use their sketches in the subsequent 3D modeling and rendering process. In this paper, we detail the core design concepts and implementation of Feather, focusing on the 3D guide and the view-oriented joystick widget, and discuss the application and potential of the 3D creative process facilitated by Feather.

2 RELATED WORK

3D sketching techniques, utilizing intuitive sketching input to generate 3D shapes, have been researched and developed for industries such as automotive [Bae et al., 2008], architecture [Dorsey et al., 2007], and product design [Kim and Bae, 2016, Kim et al., 2018]. These techniques have also been applied to virtual reality (Gravity Sketch) and animation (Blender Grease Pencil).

We combine multi-view sketching [Bae et al., 2008] with sketching surfaces [Kim and Bae, 2016, Kim et al., 2018]. Our interface transforms strokes from one viewpoint into a sketching surface and projects strokes from other viewpoints onto it, generating 3D curves. We implemented the system with web graphics interworking with cloud computing, similar to the latest cloud-based 2D design (Figma) or 3D modeling (Spline) platforms.

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3 FEATHER 3D SKETCHBOOK

Feather is a 3D sketching system that allows users to draw 3D curves with different shapes, colors, and thicknesses. To streamline the complex 3D tasks, we combined view switching and the familiar 2D sketching interface for creating and manipulating 3D curves.

3.1 Drawing & Bending 3D Guide

To generate a 3D curve, the unit element of Feather artwork, users first create a "3D guide" (Figure 2). When drawing the initial stroke, a surface as a 3D guide extrudes in the viewing direction (Figure 2a). Subsequent strokes, drawn from different viewpoints, are projected onto the guide, resulting in curves with 3D coordinates.

The concept of the 3D guide stems from the multi-view 3D sketching technique, where 2D input from two or more viewpoints defines 3D coordinate values. Since depth information is not determined in the initial view, the 3D guide serves as a geometric and visual representation of the user's intended 3D curve. We objectify this intermediate state, like a "bent paper," allowing users to examine the shape from other viewpoints and apply additional adjustments similar to existing modeling systems.

The bending function (Figure 2b) deforms the 3D guide surface to follow an additional stroke from another viewpoint while preserving the shape of the first stroke. For instance, a 3D guide created by drawing a circle with the first stroke from a frontal viewpoint would be a straight cylinder. Then if the user draws a larger circle from a top viewpoint with the bending function enabled, the 3D guide would be warped from the cylinder to a doughnut shape.

Depending on how much the user adds information from multiple viewpoints, the 3D guide can be used as an intermediary to build up 3D curves with two-view 3D sketching, or it can be used as a swept surface similar to traditional 3D modeling techniques.

3.2 View-oriented Joystick Widget

The view-oriented joystick widget is a 2D interface that enables users to translate, rotate, and scale 3D curve objects (Figure 3). It comprises a central crosshair and a translation stick in the corner surrounded by a rotation wheel and scaling handles (Figure 1). The widget's key features are that the object's translation is perpendicular to the view, and the axis of rotation and scaling is fixed at the screen center and parallel to the viewing direction.

Like drawing a 3D guide in one view and then drawing a curve in another, users utilize the widget by repeating 2D planar movements at multiple viewpoints to perform a 3D translation (Figure 3a). Unlike conventional widgets that restrict object translation to fixed X-Y-Z axes, our widget enables users to translate curves in subtle directions by changing viewpoints. Since using our widget on X-Y-Z views yields the same result as the conventional widgets, users can still quickly place 3D curves with just two viewpoint maneuvers.

Users can set the center and direction of rotation just by aiming the viewport at curves (Figure 3b). Since the rotation axis is always centered on the screen, users do not have to worry about unexpectedly rotating curves with an off-screen center and radius.

The scaling also uses the screen's center as an anchor, eliminating the need to translate after scaling (Figure 3c). For example, if the user positions the airplane's body at the screen center, scaling up the wings will only stretch them outward from the body.

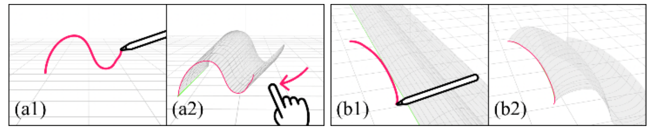

Figure 2: A first stroke (a1) creates a 3D guide (sketching surface) extruded along the first view (a2). A bending stroke (b1) warps the 3D guide from a different view (b2).

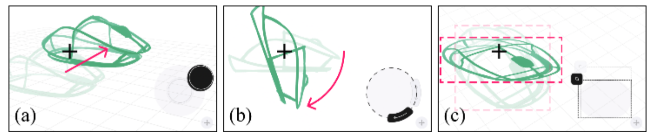

Figure 3: 3D curves are translated by the stick (a), rotated by the wheel (b), and scaled by the corner handle (c).

3.3 Cross-device Web Application

Feather is implemented with Javascript and WebGL that runs on top of a web browser and is available on the web (www.feather.art). Users can continue working with 3D curve artworks synchronized in the cloud across devices, including mobile and desktop.

4 CONCLUSION AND FUTURE WORK

To lower the barrier to first-time 3D expression of ideas, we developed Feather, which allows users to create 3D curve artwork using only 2D input for navigation and object manipulation. Extruding 3D guides with 2D sketch input could be extended to include additional surface creation techniques, such as loft and sweep. The view-oriented joystick, which operates based on direction and center of viewing, could also be used consistently for rigging and animation [Lee et al., 2022] of sketch parts. The cloud web app will enable multi-user collaboration and integration with creative technologies such as generative AI and Blender. We envision Feather facilitating the expression of ideas in the design and entertainment production and wherever 3D creation is needed.

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